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#include <iostream>
#include <string>
#include <map>

using namespace std;

class TrieNode {
public:
    bool isEndOfWord;
    map<char, TrieNode*> children;

    TrieNode() : isEndOfWord(false) {}
};

class Trie {
public:
    TrieNode *root;

    // Construct a Trie
    Trie() {
        root = new TrieNode();
    }

    // Search trie for key, return true if exist, false if not
    bool search(string key) {
        TrieNode *current = root;
        for (char ch : key) {
            if (current->children.find(ch) == current->children.end())
                return false;
            current = current->children[ch];
        }
        return current->isEndOfWord;
    }

    // Insert value into trie
    void insert(string value) {
        TrieNode *current = root;
        for (char ch : value) {
            if (current->children.find(ch) == current->children.end())
                current->children[ch] = new TrieNode();
            current = current->children[ch];
        }
        current->isEndOfWord = true;
    }

    // Display trie in pre-order
    void preorder() {
        preorderHelper(root, "", 0);
    }

private:
    void preorderHelper(TrieNode *node, string prefix, int level) {
        if (node->isEndOfWord) {
            // Print spaces based on the level
            for (int i = 0; i < level * 2; i++)
                cout << " ";
            cout << prefix << endl;
        }

        for (auto &child : node->children) {

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        preorderHelper(child.second, prefix + child.first, level + 1);
    }
};

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int main() {
    Trie *trie = new Trie();
    string command, key, value;
    while (1) {
        cin >> command;
        if (command == "insert") {
            cin >> value;
            trie->insert(value);
        } else if (command == "search") {
            cin >> key;
            if (trie->search(key))
                cout << "exist" << endl;
            else
                cout << "not exist" << endl;
        } else if (command == "print") {
            trie->preorder();
        } else if (command == "exit") {
            break;
        }
    }
    delete trie;
    return 0;
}

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